Kabelo Makgae

0664384076 • makgaekabelomotshabi.kb@gmail.com • Pretoria, GP

https://linkedin.com/in/kabelo-makgae-37122b270

Summary

Data Reliability Professional with 3+ years of experience designing and scaling secure data infrastructure for mission-critical applications. Expert in Python, PostgreSQL, and distributed systems, with a proven ability to automate database lifecycles and strengthen system resilience. Dedicated to reducing operational overhead and maintaining high availability for global payment and security-focused platforms through cloud-native architectures and advanced monitoring tools.

Work Experience

Data Science Intern *June 2026 – Present*

HEX SOFTWARES PVT. LTD., REMOTE

- Build an end-to-end Loan Eligibility Prediction model using Python and Scikit-learn to automate risk assessment and improve database reliability.
- Perform comprehensive data cleaning, preprocessing, and exploratory data analysis (EDA) to ensure data governance and consistency across distributed datasets.
- Train and evaluate multiple machine learning models including Random Forest and KNN to optimize predictive accuracy for high-stakes financial data.
- Achieve 80% model accuracy using Logistic Regression and document all workflows to strengthen reproducibility and team collaboration.
- Use Git and GitHub for version control and project documentation to maintain high standards of software configuration and code reliability.

Full Stack Developer & Data Engineer *September 2025 – March 2026*

ADDRESSME, JOHANNESBURG, ZA

- Designed scalable backend services and data pipelines using Python and FastAPI to support mission-critical payment infrastructure and federated data sharing.
- Optimized system performance and automated database lifecycle tasks to reduce compute costs and resource utilization by 25% in high-demand environments.
- Automated robust ETL workflows and data mining processes to enable real-time analytics and strengthen system reliability for intelligence-driven decision making.
- Strengthened software architectures for national security applications by collaborating with cross-functional teams in an Agile environment to oversee deployment and Prometheus-based monitoring.

Software Engineer & Data Scientist *January 2023 – January 2025*

INDEPENDENT / PROJECT-BASED WORK, REMOTE

- Engineered and scaled machine learning models using Scikit-Learn and R, achieving 70% optimization in predictive modeling for complex distributed systems.

- Analyzed large-scale, fragmented datasets using advanced statistical modeling to extract actionable intelligence and support data governance initiatives.
- Scaled analytic workflows by engineering cloud-native data solutions on AWS using Terraform and Docker to automate storage and compute optimization.
- Strengthened the security of financial datasets by applying feature engineering and predictive analytics while adhering to data privacy and PCI DSS awareness.
- Communicated complex mission analytics to partners by integrating interactive dashboards with Tableau, Power BI, and Plotly Dash.

Education

Professional Certificate in Data Science *August 2025*
COURSERA, UNITED STATES

Data Science in Applied Data Science *July 2025*
WORLDQUANT UNIVERSITY, NEW ORLEANS

Software Engineering in Software Engineering *February 2025*
ALX SOFTWARE ENGINEERING ACADEMY, SOUTH AFRICA

Electrical Engineering Technology in Electrical Engineering Technology *January 2019*
PRETORIA TECHNICAL HIGH SCHOOL, PRETORIA, SOUTH AFRICA

Additional Skills

Database Management & Reliability: PostgreSQL, SQL Server, MySQL, Azure Cosmos DB, ClickHouse, Delta Lake, SQLite, Database Reliability, Database Lifecycle Management

Cloud, Infrastructure & Monitoring: Terraform, Kubernetes, Docker, Azure, AWS (Lambda, S3), Prometheus, Cloud-Native Architecture, Distributed Systems

Data Engineering & Analytics: Python, ETL Pipeline Development, Databricks, dbt, MLflow, Data Governance, PCI DSS, Data Mining, Star Schema

Software Engineering & Tools: FastAPI, Agile, Git, R, Machine Learning, Tableau, Power BI, Plotly Dash

Certifications

- Gen AI - COURSES (March 2026)

Languages

- R - Basic

Projects

Bank Fraud Detection System

Built an end-to-end fraud detection pipeline using Databricks, Delta Lake, Docker, and MLflow. Engineered transaction-based and behavioral features for predictive risk analysis. Evaluated machine learning models including Random Forest, XGBoost, LightGBM, CatBoost, and Autoencoders. Improved fraud detection accuracy while supporting scalable analytics and monitoring workflows.

Technologies: Databricks, Delta Lake, Docker, MLflow, Random Forest, XGBoost, LightGBM, CatBoost, Autoencoders

Financial Volatility Forecasting

Developed a GARCH-based financial forecasting model using Python and statistical time-series analysis. Processed financial datasets through API integration and SQLite workflows. Built a FastAPI service for delivering predictive forecasting outputs and analytical insights.

Technologies: Python, GARCH, API, SQLite, FastAPI

Customer Segmentation & Business Intelligence

Applied clustering algorithms and PCA techniques to analyze customer behavior patterns. Developed interactive dashboards using Plotly Dash and Tableau for business reporting. Performed data preprocessing, feature engineering, and visualization using pandas and scikit-learn.

Technologies: PCA, Plotly Dash, Tableau, pandas, scikit-learn

A/B Testing & Statistical Analysis

Conducted A/B testing and chi-square statistical analysis to evaluate user engagement. Automated reporting and ETL workflows using Python and pandas. Delivered actionable insights through statistical analysis and interactive reporting systems.

Technologies: Python, pandas

AWS Serverless ETL Pipeline

Built an end-to-end ETL pipeline using AWS Lambda, S3, PostgreSQL, Docker, and dbt. Implemented automated ingestion and transformation workflows for analytics-ready reporting. Applied star schema data modeling and cloud-native architecture principles.

Technologies: AWS Lambda, S3, PostgreSQL, Docker, dbt